5.2 Absolute maximum ratings

Stresses above the absolute maximum ratings listed in Table 14, Table 15, and Table 16 may damage permanently the device. These are stress ratings only and the functional operation of the device at these conditions is not implied. Exposure to maximum rating conditions for extended periods may affect device reliability. Device mission profile (application conditions) is compliant with JEDEC JESD47 qualification standard. Extended mission profiles are available on demand.

Table 14. Voltage characteristics

All main power (VDD) and ground (VSS) pins must always be connected to the external power supply, in the permitted range.
The I/O structure options listed in this table can be a concatenation of options including the option explicitly listed. For instance, TT_a refers to any TT I/O with _a option. TT_xx refers to any TT I/O and FT_xx refers to any FT I/O.

Symbols	Ratings	Min	Max	Unit
V_{DDX} - V_{SS}	External main supply voltage (including V_{DD}, V_{DDA} and V_{REF+})	−0.3	4.0	V
V_{IN}(1)	Input voltage on FT_xxx pins	VSS−0.3	MIN (V_{DD} + 4.0V , 6.0 V)(2)	V
	Input voltage on TT_xx pins	VSS−0.3	4.0	V
V_{REF+}-V_{DDA}	Allowed voltage difference for $V_{REF+} > V_{DDA}$	-	0.4	V
$\|\Delta V_{DDX}\|$	Variations between different VDDX power pins of the same domain	-	50.0	mV
|VSSx-VSS|	Variations between all the different ground pins	-	50.0	mV

1. *V_{IN} maximum must always be respected. Refer to Table 15 for the maximum allowed injected current values.*
2. *When the analog option is selected by enabling analog peripheral or the pull-up/pull-down resistors are enabled on a given pin, V_{IN} must not exceed 4 V.*

Table 15. Current characteristics

Symbol	Ratings	Max	Unit
$\sum IV_{DD}$	Total current into sum of all V_{DD} power lines (source)(1)	200	mA
$\sum IV_{SS}$	Total current out of sum of all V_{SS} ground lines (sink)(1)	200	
IV_{DD}	Maximum current into each VDD power pin (source)(1)	100	
IV_{SS}	Maximum current out of each VSS ground pin (sink)(1)	100	
I_{IO}	Output current sunk by any I/O and control pin	20	
	Output current sourced by any I/O and control pin	20	
$\sum I_{(PIN)}$	Total output current sunk by sum of all I/Os and control pins(2)	140	
	Total output current sourced by sum of all I/Os and control pins(2)	140	
$I_{INJ(PIN)}$(3)(4)	Injected current on FT_xx, TT_xx, RST pins	-5/0	
$\sum I_{INJ(PIN)}$	Total injected current (sum of all I/Os and control pins)(5)	±25	

1. *All main power (V_{DD}) and ground (V_{SS}) pins must always be connected to the external power supplies, in the permitted range.*
2. *This current consumption must be correctly distributed over all I/Os and control pins. The total output current must not be sunk/sourced between two consecutive power supply pins, referring to high pin count QFP packages.*
3. *A negative injection is induced by $V_{IN} < V_{SS}$. $I_{INJ(PIN)}$ must never be exceeded. Refer also to Table 14 for the minimum allowed input voltage values.*
4. *Positive injection (when $V_{IN} > V_{DDIOx}$) is not possible on these I/Os and does not occur for input voltages lower than the specified maximum value.*
5. *When several inputs are submitted to a current injection, the maximum $\sum I_{INJ(PIN)}$ is the absolute sum of the negative injected currents (instantaneous values).*